

RX513 Progress Note Form: Vancomycin Initiation

PATIENT IDENTIFYING INFORMATION (MR#)

STATEMENT OF REASON FOR CONSULT: Marty Byrde is a 54 year old male with newly diagnosed AML undergoing induction chemotherapy (currently day 3) who developed a fever, cough, and shortness of breath. Pharmacy consulted to dose vancomycin and provide additional recommendations for neutropenic fever.

S/O: (SUBJECTIVE & OBJECTIVE)

Past Medical History relative to ID or resistance risk factors: AML induction chemotherapy with fever on day 3. DVT (2 months ago), HTN, T2DM, motor vehicle accident 5 months ago

Vital signs: Ht: 6'1", Wt: 97 kg, IBW: 79.9kg,
BP: 102/68mmHg, P 95bpm, RR 18, Pulse ox 89% temp: 38.8°C

Laboratory Data:

BUN 21/Scr 0.8mg/dL, Albumin 2.8g/dL, WBC 495, ANC: 108, Plt 70 x 10³/mm³

PICC line cultures (date): pending

Blood Cultures (date): pending

Physical Exam Findings /Radiographic Imaging/Procedure and/or Diagnostic Tests:

ROS: in acute distress, chills, chest pain

PE: respiratory - wheezes/crackles; CV – chest pain

Imaging: Chest Xray: bilateral lower lobe infiltrates

Fever/chills, nausea, no cough or diarrhea, no erythema or induration around port, alert and oriented x 3.

Medication List:

Warfarin, lisinopril, metoprolol, metformin, oxycodone, oxycodone/acetaminophen, daunorubicin/cytarabine (7+3 induction), allopurinol, ceftriaxone, fluconazole, acyclovir

Allergy: Morphine

A/P: (ASSESSMENT & PLAN)

Patient has high risk febrile neutropenia (inpatient at fever onset, anticipated neutropenia > 7 days, pneumonia, MASCC score 16) and meets criteria for vancomycin (hypotension, possible pneumonia). Antimicrobial coverage should be broadened to cover pseudomonas.

Goals of therapy: treat source of infection, resolve fever and signs/symptoms of infection

Calculated pharmacokinetic parameters:

estimated CrCl ~ 119 ml/min

ke = **0.103 hr⁻¹**

t_{1/2} = **6.7 hr**

Vd = **67.9 L**

Est Cpk = **30.5mcg/ml**

Est Ctr = **10.9mcg/ml**

Est AUC = **463.4/L*hr**

Based on patient's age, weight, renal function and population pharmacokinetic parameters recommend vancomycin maintenance dose of 2000 mg every 12 hours to achieve an AUC of 400-600 mg*hr/L. If cultures without resistant gram positive organisms or s/sx of a gram positive infection, consider discontinuation of vancomycin after 48 to 72 hours. Recommend discontinuing ceftriaxone and broaden coverage to include *Pseudomonas sp.* With Antipseudomonal

beta-lactam or carbapenem (cefepime 2 g IV every 8 hours, piperacillin/tazobactam 4.5 g IV every 6 hours, imipenem/cilastatin 500 mg IV every 6 hours, meropenem 1 g IV every 8 hours) Additionally recommend:

- Continue acyclovir prophylaxis.
- Broaden fungal prophylaxis to posaconazole delayed release tablets 300 mg PO twice daily x 2 doses then 300 mg PO daily
- Hold metoprolol and lisinopril and restart once hypotension resolved.
- Monitoring: vital signs every 8 hours, draw blood cultures every 24-48 hours if fever persists, CBC with differential daily, BUN/Scr daily, s/sx of infection daily, adverse drug effects daily (beta-lactam rash, vancomycin infusion reaction, etc), vancomycin level if continued >48-72 hours.

Pharmacy will continue to follow and monitor the patient's progress, including culture and susceptibility data and adverse drug reactions. If necessary, will monitor vancomycin serum concentrations as clinical course is determined.

SIGNATURE & CONTACT INFORMATION:

Student Name, PharmD Candidate, Butler University

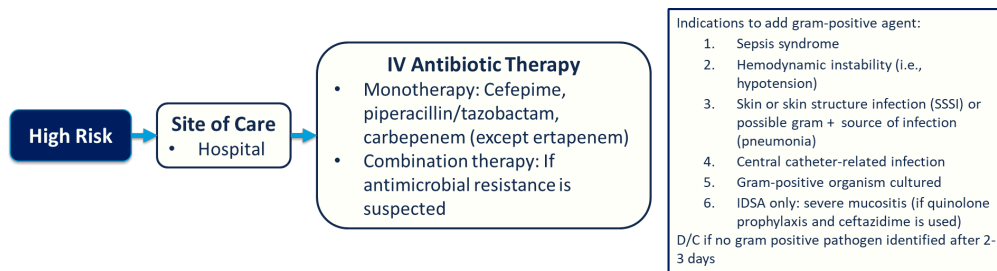
Talking points:

Low Risk	High Risk
• Outpatient status at fever onset • No acute comorbid illness requiring inpatient treatment • Anticipated duration of severe neutropenia (≤100 cells/mcL) for < 7 days • Good performance status • No hepatic or renal insufficiency • MASCC Risk-index score ≥ 21 or CISNE <3	• MASCC Risk-index score <21 or CISNE ≥ 3 • Inpatient status at fever onset • Significant medical comorbidity • Clinically unstable • Allogenic HSCT • Anticipated severe neutropenia ≥ 7days • LFTs > 5 x ULN • CrCl < 30 ml/min • Uncontrolled/progressive cancer • Pneumonia or other complex infection at clinical presentation • Severe mucositis

MASCC Risk-Index Score	
Burden of illness	
No/mild symptoms	5
Moderate symptoms	3
No Hypotension	5
No COPD	4
Solid tumor or hematologic malignancy with no previous fungal infection	4
No dehydration	3
Outpatient status	3
Age < 60 years	2
Low risk	≥21
High risk	<21

CISNE Score	
ECOG PS ≥2	2
Stress-induced hyperglycemia	2
COPD	1
Chronic cardiovascular disease	1
Mucositis	1
Monocytes < 200/μL	1
Low risk	<3
High Risk	≥3

MASCC for this patient – moderate sx (3), no COPD (4), no prior fungal infection (4), No dehydration (3), Age <60 (2) = 16



Other Info:

- Treatment course: Induction chemotherapy with 7+3 (7 days of cytarabine + 3 days of daunorubicin). Likely to be neutropenic x 4 weeks. At day 15, a bone marrow biopsy will be done to direct further anticancer treatment.
- This patient received a few days of hydroxyurea to decrease white count (leukemic cells) while waiting for diagnosis.
- No indication for growth factor products (myeloid growth factors generally contraindicated – myeloid malignancy)
- Febrile neutropenia definition: Neutropenia PLUS single PO temp > 101°F (38.3°C) or ≥ 100.4°F (38°C) for at least an hour
- This patient is high risk! Treat inpatient with IV antibiotics. If low risk, may treat outpatient (optimal regimen: Augmentin/ciprofloxacin)
- Duration of broad-spectrum antibiotics: multiple paths!
 - o Patient becomes afebrile without any identified infection and patient is stable:
 - Traditional method: continue until no longer neutropenic
 - Emerging method: may consider stepping down to quinolone prophylaxis (levofloxacin or ciprofloxacin)
 - o Patient remains febrile
 - If they are stable – continue current antibiotics; consider adding antifungal after 4 days of being febrile
 - If they are unstable – broaden coverage
 - o Patient becomes afebrile with identified infection and remains neutropenic
 - Maintain broad spectrum coverage for at least as long as a course of therapy for the site of infection would call for
 - After a full treatment course based on site of infection, could step down to prophylactic regimen (or could continue until no longer neutropenic)
 - o Patient becomes afebrile with identified infection and neutropenia resolves
 - May narrow regimen based on site of infection and continue for a duration appropriate for the site of infection

Vancomycin dosing:

1. Calculate IBW
 - a. $IBW = 50 + (2.3 \times 13) = 79.9\text{kg}$
 - b. Adjusted BW not necessary as $BMI \leq 30\text{m}^2$ (patient BMI = 28.2)
2. Calculate CrCl:
 - a. $[(140-54) \times 79.9] / (72 \times 0.8) = 119.3 \text{ ml/min}$ (round to ~ 119 ml/min)
3. Calculate elimination rate constant (k)
 - a. $k(\text{hr}^{-1}) = 0.00083 \times (\text{CrCl}) + 0.0044 = 0.103 \text{ hr}^{-1}$
4. Calculate half-life
 - a. $t_{1/2} = 0.693/k = 6.7 \text{ hr}$
5. Calculate Vd (*use ABW here)
 - a. $0.7 \times \text{ABW} = 67.9 \text{ L}$
6. Calculate vancomycin clearance
 - a. $\text{Cl}_{\text{vanc}} = k \times V_d = 0.103 \text{ hr}^{-1} \times 67.9\text{L} = 7\text{L/hr}$
7. Calculate total daily dose based on AUC goals
 - a. $\text{TDD} = \text{Cl}_{\text{vanc}} \times \text{desired AUC}$ (400-600 mg*hr/L) – we will **target 600** for these calcs.
 - b. $7\text{L/hr} \times 600 = 4200\text{mg}$
8. Dosing interval (tau)
 - a. **CrCl ≥ 60 : q12hrs**
 - b. CrCl 25-59: q24hrs
 - c. CrCl $\leq 25 \text{ ml/min}$: once
9. Calculate maintenance dose
 - a. $\text{MD} = \text{TDD} \times [\text{tau}/24 \text{ hrs}] = 4200 \times (12/24) = 2100 \text{ mg}$. **Decide how to round... 2000mg or 2250mg .**
 - b. Discussion: We used 600 as AUC target – so if increase dose – what would this do to AUC?? Therefore, would not round up due to likelihood of overshooting target.
10. Calculate expected peak:
 - a. **Using 2000mg q 12hr**
 - i. $[\text{MD} \times (1 - e^{(-k \times \text{tinf})}) / (e^{-k \times T})] / [k \times V_d \times T_{\text{inf}} \times (1 - e^{(-k \times \text{tau})})]$
 - ii. $\text{tinf}: 2 \text{ h}; T: 2\text{h}$
 - iii. $[2000 \times (1 - e^{(-0.103 \times 2)}) / (e^{-0.103 \times 2})] / [0.103 \times 67.9 \times 2 \times (1 - e^{(-0.103 \times 12)})]$
 - iv. **Peak = 30.5mcg/ml**
11. Calculate expected trough
 - a. **Using 2000mg q 12 hrs**
 - i. $\text{Peak} \times e^{(-k \times (\text{tau} - 2))}$
 - ii. $30.5 \times e^{(-0.103 \times (12 - 2))}$
 - iii. **Trough = 10.9mcg/ml**
12. Calculate AUC infusion
 - a. $\text{AUC} = [(\text{Peak} + \text{trough})/2] \times \text{tinf}$
 - b. $\text{AUC} = [(30.5 + 10.9)/2] \times 2 = 41.4\text{mg/L} \times \text{hr}$
13. Calculate AUC elimination
 - a. $[(\text{Peak} - \text{trough})/k]$
 - b. $[(30.5 - 10.9)/0.103] = 190.3\text{mg/L} \times \text{hr}$
14. Calculate 24 hour AUC
 - a. $[(\text{AUC inf} + \text{AUC elim}) \times 24/\text{tau}]$

b. $[(41.4+190.3) \times 24/12] = 463.4 \text{ mg/L*hr}$ (within goal of 400-600mg/L*hr)

15. Determine Final Dose recommendation

a. **2000mg IV q 12 hrs**

Per Damer – taught to keep 3 significant figures throughout work. Damer teaches to round to 10th place for time or volume. Ke - thousandths